Matoshri College of Engineering and Research Centre

Department of Engineering Sciences

Presentation on

QUANTUM MECHANICS

SUPERCONDUCTORS AND NANOTECHNOLOGY

By

Mansi Subhash Zalte

Roll No.:60

Under The Guidance of
Ms.Reshma Kadale Mam

Course Name: Applied Physics & Chemistry

Academic Year: 2025-26

Div-H | Sem-II | Applied Physics & Chemistry

ADVANCE LEARNER
ASSIGNMENT

Exploring Quantum Mechanics, Superconductors & Nanotechnology

Wave-Particle Duality

Heisenberg's Uncertainty

Superconductors

Carbon Nanotubes

OVERVIEW

QUANTUM MECHANICS

- Wave-Particle Duality
- de Broglie's Hypothesis
- Double-Slit Experiment
- Heisenberg's Uncertainty Principle
- Consequences & Applications

SUPERCONDUCTORS & NANOTECHNOLOGY

- High-Temperature Superconductors
- Challenges in Superconductivity
- Carbon Nanotubes (CNTs)
- Properties of CNTs
- Applications

WAVE-PARTICLE DUALITY

Quantum Mechanics — Topic 1

Definition: Matter and energy exhibit both wave-like and particle-like properties, depending on how they are observed.

de Broglie's Hypothesis

Every moving particle has an associated wavelength: $\lambda = h/mv$, where h is Planck's constant.

Photoelectric Effect

Light behaves as a stream of photons (particles) that eject electrons from metal surfaces.

Double-Slit Experiment

Electrons fired at a double slit produce an interference pattern — proof of wave nature.

Compton Scattering

X-rays scatter off electrons with a shift in wavelength, confirming photon momentum.

Key Equation: $\lambda = h / mv$ (de Broglie wavelength) | $E = hf$ (Planck's energy-frequency relation)

HEISENBERG'S UNCERTAINTY PRINCIPLE

Quantum Mechanics — Topic 2

Principle: It is impossible to simultaneously determine the exact position and exact momentum of a subatomic particle.

$$\Delta x \cdot \Delta p \geq \hbar / 2$$

No Classical Orbits

Electrons don't follow fixed paths around the nucleus; they exist in probability clouds called orbitals.

Zero-Point Energy

Quantum particles always possess some minimum energy and can never be completely at rest.

Quantum Tunneling

Particles can penetrate energy barriers they classically shouldn't cross, enabling nuclear fusion in stars.

Electron Microscopy Limits

Observing a particle disturbs it; using high-energy photons to see small objects changes their momentum.

HIGH-TEMPERATURE SUPERCONDUCTORS

Superconductors & Nanotechnology — Topic 1

Superconductors are materials that conduct electricity with zero electrical resistance below a critical temperature (Tc). High-Tc superconductors operate above 77 K (liquid nitrogen temperature).

Key Features

- Zero electrical resistance below Tc
- Meissner Effect: expels magnetic fields
- Cooper pairs of electrons carry current
- Example: YBCO (Tc ≈ 92 K), BSCCO (Tc ≈ 110 K)

Challenges

- Brittle ceramic materials — hard to fabricate
- Cooling still needed (costly cryogenics)
- Mechanism not fully understood theoretically
- Limited to low current densities in some cases

Applications: MRI machines | Maglev trains | Particle accelerators (LHC) | Quantum computing

CARBON NANOTUBES & THEIR PROPERTIES

Superconductors & Nanotechnology — Topic 2

Carbon Nanotubes (CNTs) are cylindrical nanostructures of carbon atoms arranged in a hexagonal lattice, with diameters of 1–100 nm. They can be single-walled (SWCNT) or multi-walled (MWCNT).

Mechanical Strength

~100× stronger than steel

Young's modulus ~1 TPa

Electrical Conductivity

Metallic or Semiconducting

Depends on chirality/rolling angle

Thermal Conductivity

~3500 W/m·K

Higher than diamond

Low Density

~1.3–1.4 g/cm^3

Extremely lightweight material

High Aspect Ratio

Length >> Diameter

Enables nanoscale engineering

Chemical Stability

Inert in most environments

Resistant to corrosion & oxidation

APPLICATIONS OF CARBON NANOTUBES

Electronics

- Transistors smaller than silicon
- Nano-scale wires & interconnects
- Field-emission displays

Medicine & Biology

- Drug delivery systems
- Biosensors for disease detection
- Scaffold for tissue engineering

Energy

- Supercapacitors & batteries
- Hydrogen storage
- Solar cell electrodes

Composites & Materials

- Ultra-strong composite materials
- Lightweight aerospace structures
- Wear-resistant coatings

SUMMARY

Key Takeaways

01

Wave-Particle Duality

Light and matter exhibit both wave and particle properties, forming the foundation of quantum theory.

02

Uncertainty Principle

Heisenberg proved fundamental limits to simultaneous measurement, shaping modern quantum mechanics.

03

Superconductors

High-Tc materials offer zero resistance with wide applications; challenges remain in fabrication.

04

Carbon Nanotubes

CNTs possess extraordinary strength, conductivity, and thermal properties for next-gen technology.